

*MA.6.NSO.4.1***Benchmark**

MA.6.NSO.4.1 Apply and extend previous understandings of operations with whole numbers to add and subtract integers with procedural fluency.

Benchmark Clarifications:

Clarification 1: Instruction begins with the use of manipulatives, models and number lines working towards becoming procedurally fluent by the end of grade 6.

Clarification 2: Instruction focuses on the inverse relationship between the operations of addition and subtraction. If p and q are integers, then $p - q = p + (-q)$ and $p + q = p - (-q)$.

Connecting Benchmarks/Horizontal Alignment

- MA.6.NSO.1
- MA.6.AR.1.3
- MA.6.AR.2.2
- MA.6.GR.1.2, MA.6.GR.1.3
- MA.7.NSO.2.2

Terms from the K-12 Glossary

- Equal Sign
- Equation
- Integers
- Number Line
- Whole Number

Vertical Alignment**Previous Benchmarks**

- MA.3.NSO.2.1

Next Benchmarks

- MA.8.NSO.1.7

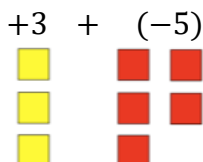
Purpose and Instructional Strategies

In previous courses, students added and subtracted whole numbers using models, including number lines and counters. These skills are extended in grade 6 accelerated to compare and perform operations with integers with procedural fluency as well as write and solve single-step equations using integers. Students also perform all operations with rational numbers in the form of expressions, equations and inequalities. In future courses, students will use the order of operations with rational numbers, including exponents and radicals to solve multi-step mathematical and real-world problems.

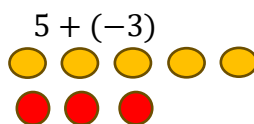
- It can be beneficial for students to learn to add and subtract integers using real-world contexts, such as, the movement of a football on a football field, above and below sea level, temperature; refer to Appendix A for more situations. Given a real-world context, develop a conceptual model to represent the situation. Students can describe the relationship mathematically as an expression.
- Explore opportunities with students to discover patterns or rules instead of being directly instructed and expected to memorize them. It is not expected or recommended that students start by generating equivalent expressions mathematically (*MTR.2.1, MTR.5.1, MTR.7.1*).

- Use models to help students identify patterns for adding and subtracting integers. Instruction includes opportunities to model adding and subtracting integers using tools such as algebra tiles, two-color counters and number lines. Students should relate the model to the expression (*MTR.5.1*).

- Algebra Tiles

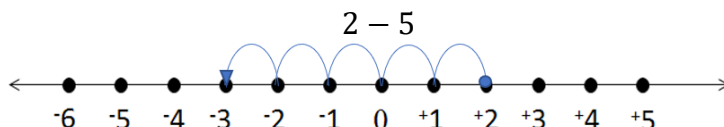


- Two-Color Counters



- For example, if a student is evaluating $5 + (-3)$ using two-colored counters, a student can think, “I start with 5 yellows and 3 reds. A positive and a negative together make a zero pair. So, if I have 5 positive yellows and 3 negative reds, 3 of the paired chips will have a sum of zero, and I will have 2 yellow positives remaining, therefore my answer is 2.”

- Number Lines



- For example, if a student is evaluating $2 - 5$ using a number line, a student can think, “I start at positive 2 and move 5 in the negative direction (to the left). I am now at -3.” Students can also use the commutative property to change the order to $(-5) + 2$. The student can see that if they start at -5 and move 2 in the positive direction (right), they land on -3.
- When using algebra tiles or two-color counters, group the positive and negative values to create zero pairs, then the remaining tiles or counters show the resulting value.
- When combining integers on a number line, a negative symbol (“-”) denotes the direction that will be traveled. If two negative symbols “-(-)” (or subtracting a negative number) are present, the operators are showing to move in the opposite direction twice; the first negative symbol (-) indicates moving to the left on the number line, whereas the second negative symbol (-) indicates moving the opposite direction, or to the right, so it is the same as adding the second addend.
- Instruction includes understanding that subtracting a positive value is the same as adding a negative value, move the value left or decrease it on the number line. The commutative property can help students to see the relationship. Subtracting a negative value is the same as adding a positive value, moving the value to the right.

- Procedural fluency is developed with the combination of conceptual understanding and practice. Students should be encouraged to think through why a sum or difference would result in a positive or negative value instead of being encouraged to memorize a set of rules (*MTR.3.1*).
- The commutative property of addition allows for individuals to rearrange the order of the terms they are adding, regardless if they are positive or negative terms (*MTR.3.1*).
 - For example, $6 - 3$ can be rewritten as $6 + (-3)$ in order to the commutative property to see that this is equivalent to $(-3) + 6$.
- Use real-world contexts that model opposite relationships, like borrowing/earning money or falling/climbing, to help students conceptually reason through why a solution would be positive or negative (*MTR.6.1*).
- Instruction includes connecting understanding of the magnitudes of numbers to the absolute value of a number.
- Students should develop fluency with and without the use of a calculator when performing operations with integers.
- Instruction includes the use of models to demonstrate that the commutative and associative properties do not apply to subtraction. Demonstrate the correct use of the commutative and associative properties of addition by rewriting a subtraction expression as an addition expression before applying either property.
 - For example, $6 - 3 \neq 3 - 6$, but $6 - 3 = 6 + (-3)$ and $6 - 3 = (-3) + 6$.
 - For example, $(3 - 6) + 5 \neq 3 - (6 + 5)$, but $(3 - 6) + 5 = 3 + (-6) + 5$ and $(3 - 6) + 5 = 3 + (-6 + 5)$.

Common Misconceptions or Errors

- Some students may misunderstand when adding and subtracting integers, the sign of the “bigger number” is kept.
 - For example, in the expression $-8 + 4$, the sum will be negative because -8 is the “bigger number.” Instead, it can be thought of as there are more negatives than positives being combined, so the result will be a negative value.
- Students may confuse zero for being a negative number because it is less than a whole. However, 0 is a neutral number; it is neither positive nor negative.
- Students may incorrectly believe that the number with the larger absolute value will always be the minuend (the first term in a subtraction expression) or that the positive number must always be the minuend. This may stem from an overgeneralization of the common pattern in elementary mathematics that “the larger number always goes first in subtraction.”

Strategies to Support Tiered Instruction

- Teacher provides opportunities for students to utilize two-color counters to represent addition and subtraction of integers. Use repeated addition to develop generalizations for multiplying integers and discuss the difference between generalized rules for addition and generalized rules for multiplication.
- Teacher provides opportunities for students to utilize two-color counters when evaluating the sum or difference of integers. Instruction includes having students write the expression, representing the expression with the two-color counters, finding the sum or difference, and then recording the solution.

- Teacher models identifying zero pairs using manipulatives such as two-color counters.

Instructional Tasks

Instructional Task 1 (MTR.7.1)

At 9:00 a.m. the temperature was -5°F . By noon, the temperature had risen 14°F . On a number line, show how to find the temperature at noon. Explain.

Instructional Task 2 (MTR.2.1)

Jonah is a novice when it comes to scuba diving. His first dive was 12 feet deep, and his second dive was 3 feet deeper than the first. Describe the depth of Jonah's second dive. Show your work on a vertical number line.

Instructional Task 3 (MTR.2.1)

At 6:00 p.m., the temperature was 11°C . By midnight, the temperature was -3°C .

Part A. Using a visual model, show how to find the difference between the temperatures.

Part B. Write a numerical expression to represent the temperature change from 6:00 p.m. to midnight. Explain.

Instructional Task 4 (MTR.2.1)

Write $-3 - 6$ as an equivalent addition problem. How are $-3 - 6$ and the addition problem you wrote related? Explain using a visual model.

Instructional Items

Instructional Item 1

Evaluate $-5 - 8 - (-2)$.

Instructional Item 2

What is the value of the expression $9 + (-12)$?

**The strategies, tasks and items included in the BIG-M are examples and should not be considered comprehensive.*